SupportIQ — Multi-Agent Implementation Architecture

How AI Agents Built This Platform End-to-End

The Core Idea

SupportIQ was not written manually line by line. It was built using a **multi-agent AI development system** where specialist AI agents collaborated — each agent completing one job, passing its output to the next agent as structured input.

This is the same principle as a software team: an architect designs, developers implement in parallel, a tester writes tests, a reviewer approves. Except every role is an AI agent, the handoffs are automated, and the entire cycle runs in minutes.

Two Layers of Multi-Agent Usage

```
┌─────────────────────────────────────────────────────────┐
│  LAYER 1 — SDLC (How SupportIQ was BUILT)               │
│  Multi-agent workflows orchestrated the entire development   │
│  pipeline: research → design → parallel implementation →     │
│  tests → review → commit                                │
├─────────────────────────────────────────────────────────┤
│  LAYER 2 — RUNTIME (What SupportIQ DOES at production time) │
│  AI support intelligence: classify, score, triage, draft     │
│  5 AI-powered endpoints backed by EPAM DIAL             │
└─────────────────────────────────────────────────────────┘
```

This document covers **Layer 1** — the multi-agent system that built the platform.

Specialist Agents — Each With a Defined Role

Six dedicated AI agents were configured, each with a custom system prompt tuned to a specific domain:

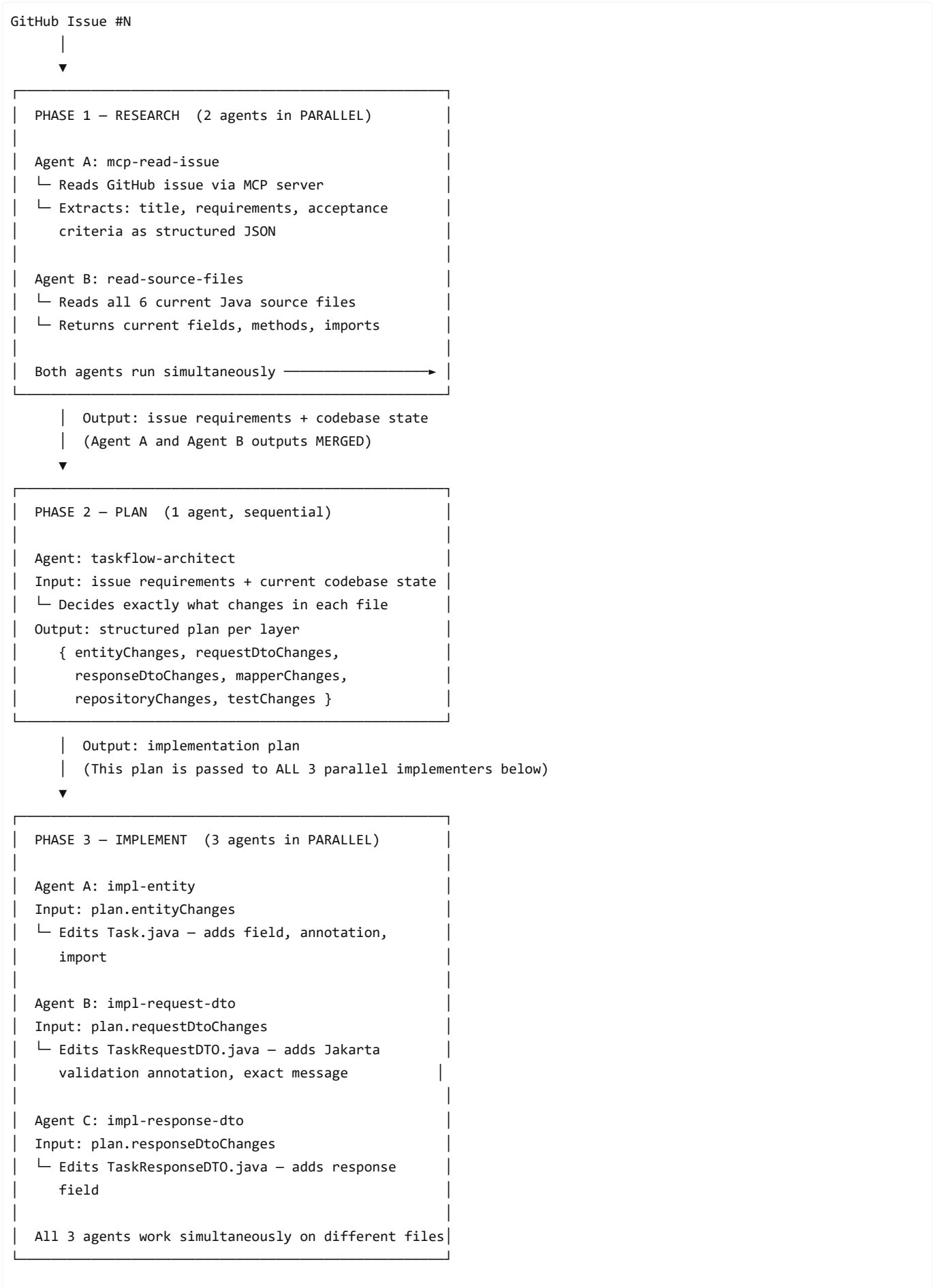
Agent	Domain	What It Does
`taskflow-architect`	System design	Reads requirements, analyses current codebase state, produces a precise implementation plan per file
`taskflow-reviewer`	Code quality	Reviews changed files against 20+ Spring Boot conventions, returns structured findings with severity ratings
`taskflow-tester`	Test engineering	Writes JUnit 5 + MockMvc tests, knows the exact `@MockBean` / `@WebMvcTest` patterns required
`support-analyst-agent`	AI/LLM design	Designed the system prompts, JSON schema contracts, and sentiment/escalation scoring logic for SupportIQ
`support-triage-agent`	Queue management	Analysed the triage strategy, SLA implications, and urgency ranking logic for the bulk triage endpoint
`demo-guide-agent`	Presentation	Produced demo scripts, management talking points, and ROI framing from real API response data

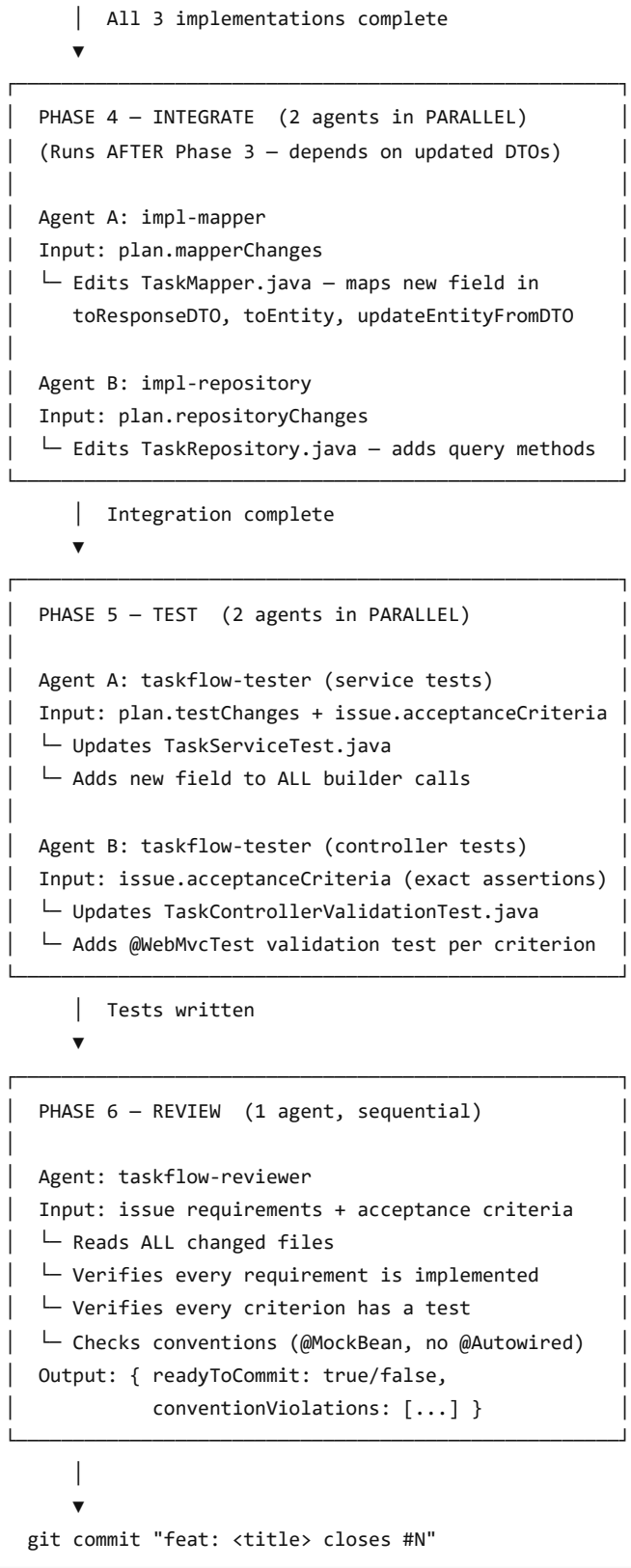
---

Workflow 1 — Feature Implementation

How a GitHub Issue Became Production Code

Every SupportIQ feature was implemented through this 6-phase automated pipeline:





Agent communication: Each phase passes its output as direct input to the next. The architect's plan flows into 5 parallel implementers. The issue's acceptance criteria flow directly into the test agent's assertions. The final reviewer receives both the original requirements AND reads the actual code — so it can verify whether they match.

Workflow 2 — Code Review

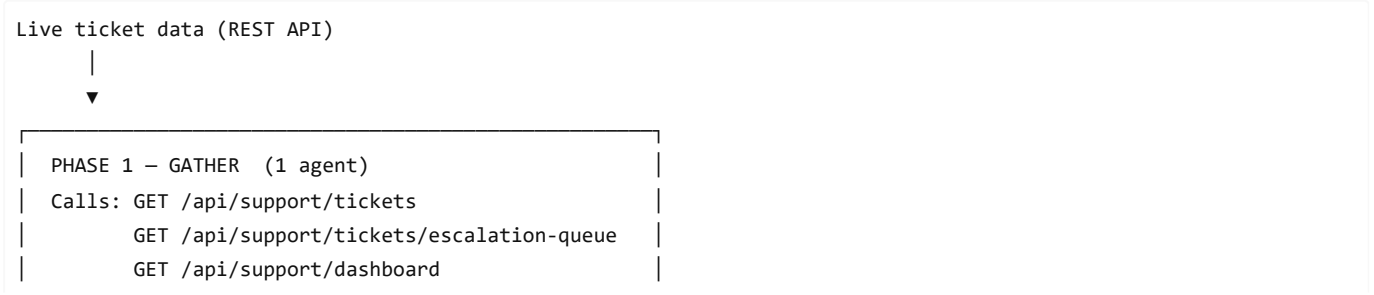
4 Specialist Agents Reviewing Simultaneously

Before any code was committed to the SupportIQ feature branch, a parallel code review workflow ran:

Workflow 3 — Support AI Insights

4 Parallel Analysts + Executive Briefing

This workflow runs against the **live SupportIQ queue** and produces a management-level intelligence briefing:



```
│  Returns: structured queue summary                  │
└─────────────────────────────────────────────────────┘
      │  Queue data
      │  (Same data passed to ALL 4 analysts below)
      ▼
┌─────────────────────────────────────────────────────┐
│  PHASE 2 — ANALYZE  (4 agents in PARALLEL)          │
│  Each agent receives the SAME queue data            │
│  but analyses a different dimension:                │
│                                                     │
│  Agent 1: Queue Health & Operations                 │
│  └─ Open/resolved ratio, escalation rate,           │
│     IN_PROGRESS activity                            │
│  └─ Score: 0–100                                    │
│                                                     │
│  Agent 2: Escalation & Churn Risk                   │
│  └─ Highest churn risk customers, legal threats,    │
│     systemic patterns, financial impact             │
│  └─ Score: 0–100                                    │
│                                                     │
│  Agent 3: Customer Sentiment & Experience           │
│  └─ Emotional state of customer base, angry vs      │
│     satisfied distribution, NPS prediction          │
│  └─ Score: 0–100                                    │
│                                                     │
│  Agent 4: Issue Patterns & Root Cause               │
│  └─ Dominant categories, product/process signals,   │
│     self-service opportunities                      │
│  └─ Score: 0–100                                    │
└─────────────────────────────────────────────────────┘
      │  4 scores + 4 finding sets + urgent actions
      │  (All 4 outputs MERGED before next phase)
      ▼
┌─────────────────────────────────────────────────────┐
│  PHASE 3 — SYNTHESIZE  (1 agent, sequential)        │
│  Input: 4 dimension scores + all findings merged    │
│  └─ Writes an executive briefing a VP reads         │
│     in 2 minutes                                    │
│  Output: overallHealthScore, businessRisks,         │
│          wins, topPriorityActions                   │
└─────────────────────────────────────────────────────┘
```

How Agents "Talk" to Each Other

The agents do not communicate directly in real time. Instead they communicate through **structured data handoffs** — the output of one agent is the input of the next.

```
Agent A produces:
{
  "requirements": ["Add dueDate field", "Validate: past dates rejected"],
  "acceptanceCriteria": ["POST with past date returns 400", "Message: Due date must be..."]
}
                │
                │ This exact JSON is passed into Agent B's prompt
                ▼
Agent B (architect) reads the requirements and produces:
```

```
{
  "entityChanges": ["Add LocalDate dueDate field with @Column(nullable=true)"],
  "requestDtoChanges": ["Add @FutureOrPresent with message 'Due date must be today...'"],
  "testChanges": ["Add pastDate rejection test in controller test"]
}
                │
                │ This exact plan is passed into 3 parallel implementers
                ▼
Agent C, D, E each implement their assigned file simultaneously
```

This is structured agent communication — each agent's response is a typed contract that the orchestrator passes forward. No agent needs to "know" about the others. The workflow is the coordinator.

Results

Metric	Value
Total specialist agents	6
Workflows orchestrated	5
Parallel agent phases across all workflows	8
Test suite size	103+ tests
Lines of production Java	~2,000+
Time to build SupportIQ (all 5 endpoints + UI + agents + workflows)	Day 8 of the series

---

Key Takeaway for Architects

The multi-agent pattern used here is a **production-grade AI SDLC**, not a prototype. Every agent has:

- A defined role and system prompt
- A typed input schema (what it receives)
- A typed output schema (what it returns)
- A single responsibility — no agent does more than one job

The orchestration (what runs in parallel, what waits, what feeds forward) is deterministic JavaScript — not model-driven. The models handle reasoning within their scope; the workflow handles coordination across scopes.

This is the **AI-native engineering pattern**: use AI not just in the product, but in the process of building the product.

Multi-agent SDLC implemented using Claude Code workflows — Day 8, AI-Native Engineering Series